Teaching Machines to Listen Like People, Only Better

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The crack of a bat and the roar of a crowd can tell a baseball fan something exciting just happened, even without seeing the home run. Humans may be visual creatures, but we also learn about the world by listening. At MIT, Professor Paris Smaragdis wants to build machines that can hear and understand sound even better than humans.

A machine with super-human hearing could **pinpoint** a traffic accident by listening for squealing tires and crunching metal, or identify when an industrial drill might break while boring through dense bedrock to dig a **geothermal** well. As Smaragdis puts it, he does the same thing computer vision researchers do, but for hearing.

Smaragdis, who joined MIT in 2025, leads a research group exploring machine learning for sound source separation, smart devices that can **augment** people's audio surroundings, and better tools to help people understand, manipulate, analyze, and create music.

His path began with music. He grew up in Greece and began playing classical guitar. A turning point came when his father brought home a **synthesizer** that let him make sounds. At Berklee College of Music in Boston, he explored computers and **synthesizers** to shape sound, and majored in bass.

With mentor Richard Boulanger, he created tools to help people use computers to write, produce, and enhance music, then worked with Professor Barry Vercoe at the MIT Media Lab on software **synthesizers**.

In the Media Lab, his attention shifted to making computers understand sound. Many students then used **signal processing** to help machines handle sound, but Smaragdis chose the less explored approach of machine learning. We need to teach computers how to listen, he concluded, because we cannot simply tell them how: we lack a vocabulary to explain listening, or how to tell a cat's meow from a violin.

Word Treasure

pinpoint -- To pinpoint something is to find its exact place.

geothermal -- Geothermal means heat that comes from deep inside the Earth.

synthesizer -- A synthesizer is an instrument that can make sounds electronically.

signal processing -- Signal processing is a way of working with sounds so a machine can handle them.

augment -- To augment something is to make it better or stronger.

Background you need

Machine listening is the work of teaching computers to make sense of sounds, in the same way computer vision teaches them to make sense of images.

A computer cannot simply be told how to listen, because people do not have a ready set of words to explain how they tell one sound from another, so researchers teach machines by showing them many examples.

Smaragdis came to this work through music, first playing and shaping sounds with instruments and computers, and later turning to how machines themselves can understand sound.

His group now explores tools that separate sounds, improve the sounds around people, and help people work with music.

Break it down

LEAD: Sounds can tell people what is happening, and Smaragdis wants machines to hear even better than humans.

KEY EVENT: Smaragdis leads a research group exploring machine learning for sound.

+- WHO: Smaragdis, who works at MIT

+- WHAT: Machines that can pinpoint accidents, spot a drill about to break, and help people with music

+- WHY: Computers cannot simply be told how to listen, so they must be taught

BACKGROUND: His musical path, from classical guitar in Greece to computers and synthesizers at Berklee

EVIDENCE: With Boulanger he created tools for writing and producing music, then worked with Vercoe at the MIT Media Lab

OPEN QUESTION: How to teach computers to tell sounds apart, like a cat's meow from a violin

Why it matters

Machines that truly understand sound could warn people about danger sooner and help people hear their surroundings more clearly, from busy streets to workplaces.

Test what you remember

1. What does Smaragdis want machines to do?

(A) Play classical guitar on stage
(B) Dig geothermal wells by themselves
(C) See farther than telescopes
(D) Hear and understand sound even better than humans

2. What could super-human hearing help find?

(A) A hidden room in a school
(B) A new planet in the night sky
(C) A lost baseball in a crowd
(D) A traffic accident, from squealing tires and crunching metal

3. What did Smaragdis major in at Berklee College of Music?

(A) Drums
(B) Piano
(C) Violin
(D) Bass

4. Who was Smaragdis's mentor?

(A) His father
(B) Barry Vercoe
(C) A baseball fan
(D) Richard Boulanger

5. Why did Smaragdis say computers must be taught to listen?

(A) Because people lack a vocabulary to explain listening
(B) Because computers cannot hear at all
(C) Because guitars are hard to play
(D) Because synthesizers are too expensive

6. What does Smaragdis's research group explore?

(A) Machine learning for sound source separation
(B) Building concert halls
(C) Making violins sound like cats
(D) Designing baseball bats

Answer key (try the quiz first!): 1: D 2: D 3: D 4: D 5: A 6: A

Questions to think about

1. Paris Smaragdis: He says his work does for hearing what computer vision researchers do for sight, and that computers must be taught to listen because listening cannot simply be explained to them.

MY THOUGHTS (at least 20 words):